# Autoregressive model of order 1 (AR1)

## Parametrization

The autoregressive model of order 1 (AR1) for the Gaussian vector $\mathbf{x} = (x_1, \ldots, x_n)$ is defined as:

$$\begin{aligned} x_1 &\sim \mathcal{N}(0, (\tau(1-\rho^2))^{-1}) \\ x_i &= \rho\, x_{i-1} + \epsilon_i; \qquad \epsilon_i \sim \mathcal{N}(0, \tau^{-1}) \qquad i = 2, \ldots, n \end{aligned}$$

where

$$|\rho| < 1$$

## Hyperparameters

The precision parameter $\kappa$ is represented as

$$\theta_1 = \log(\kappa)$$

where $\kappa$ is the *marginal* precision,

$$\kappa = \tau(1-\rho^2).$$

The parameter $\rho$ is represented as

$$\theta_2 = \log\left(\frac{1+\rho}{1-\rho}\right)$$

and the prior is defined on $\theta = (\theta_1, \theta_2)$.

## Specification

The AR1 model is specified inside the `f()` function as

```
f(<whatever>, model="ar1", values=<values>, hyper = <hyper>)
```

The (optional) argument `values` is a numeric or factor vector giving the values assumed by the covariate for which we want the effect to be estimated. See the example for RW1 for an application.

### Hyperparameter spesification and default values

**doc** `Auto-regressive model of order 1 (AR(1))`

**hyper**

**theta1**

**hyperid** `14001`
**name** `log precision`
**short.name** `prec`
**prior** `loggamma`
**param** `1 5e-05`
**initial** `4`
**fixed** `FALSE`
**to.theta** `function(x) log(x)`
**from.theta** `function(x) exp(x)`

**theta2**

**hyperid** `14002`

**name** logit lag one correlation
**short.name** rho
**prior** normal
**param** 0 0.15
**initial** 2
**fixed** FALSE
**to.theta** function(x) log((1 + x) / (1 - x))
**from.theta** function(x) 2 * exp(x) / (1 + exp(x)) - 1

**theta3**

**hyperid** 14003
**name** mean
**short.name** mean
**prior** normal
**param** 0 1
**initial** 0
**fixed** TRUE
**to.theta** function(x) x
**from.theta** function(x) x

**constr** FALSE

**nrow.ncol** FALSE

**augmented** FALSE

**aug.factor** 1

**aug.constr**

**n.div.by**

**n.required** FALSE

**set.default.values** FALSE

**missing.values.warning** TRUE

**pdf** ar1

## Example

In this exaple we implement an ar1 model observed with Poisson counts

```
#simulate data
n = 100
rho = 0.8
prec = 10
## note that the marginal precision would be
marg.prec = prec * (1-rho^2)

E=sample(c(5,4,10,12),size=n,replace=T)
eta = as.vector(arima.sim(list(order = c(1,0,0), ar = rho), n = n,sd=sqrt(1/prec)))
y=rpois(n,E*exp(eta))
```

```
data = list(y=y, z=1:n, E=E)

## fit the model
formula = y~f(z,model="ar1")
result = inla(formula,family="poisson", data = data, E=E)
```

### Notes

A third hyperparameter $\theta_3$ is **experimental**, and the *mean* of the AR1 process. By default this parameter is fixed to the be zero.